UU-P-268G
October 30, 1973
SUPERSEDING
Fed. Spec. UU-P-268F
May 11, 1967 and
MIL-P-43773
19 April 1972

FEDERAL SPECIFICATION

PAPER, KRAFT, WRAPPING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for unbleached kraft wrapping paper, with or without fire resistant treatment, suitable for general wrapping purposes.

1.2 Classification.

1.2.1 Types and grades. The kraft wrapping paper covered by this specification shall be of the following types and grades, as specified (see 6.2):

Type I - Untreated

Grade A - No. 1 commercial designation.
Grade B - No. 2 commercial designation.

Type II - Fire resistant treated

Grade C - Heavy duty
Grade D - Light duty

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

MMM-A-260 - Adhesive, Water-Resistant (For Sealing Waterproofed Paper).

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PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.

Federal Standard:

FED-STD-123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards, as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications and Standards, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Technical Association of the Pulp and Paper Industry (TAPPI) Publications:

T 401 - Fiber Analysis of Paper and Paperboard.
T 403 - Bursting Strength of Paper.
T 408 - Rosin in Paper and Paperboard.

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T 410 - Weight Per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 414 - Internal Tearing Resistance of Paper.
T 461 - Flame Resistance of Treated Paper and Paperboard.

(Copies of TAPPI testing methods may be obtained from the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, Georgia 30341.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Stock. The paper shall consist of 100 percent unbleached sulfate fibers with the following exceptions: If reclaimed fibers are used (see 6.1), the finished paper may contain fibers other than unbleached sulfate (not exceeding 25 percent of the total weight of the paper stock). Reclaimed fibers in any percentage are permitted provided that the other requirements of this specification are met. Reclaimed fibers shall be from one or more of the sources listed in 6.4.

3.1.2 Sizing. The paper shall be rosin-sized.

3.2 Form. The paper shall be in sheet or roll form, as specified (see 6.2).

3.2.1 Rolls. Paper in roll form shall be evenly and compactly wound on a **removable shaft or on a paper core and shall have an inside diameter of not less than 2-3/8 inches. The length of the core shall be equal to the width of the roll with a tolerance of ± 1/4 inch. The core ends shall be positioned flush, ± 1/4 inch, with the roll faces. All rolls shall be protected with snug-fitting**

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wood, metal, or composition plugs and shall have a centered 1-1/16 ± 1/16 inch diameter hole. The end of the roll shall be secured with tape or adhesive to prevent unwinding.

3.2.2 Splices. No roll shall contain more than two splices and the distance between splices shall be not less than 20 yards. For paper in sheet form, there shall be no splices within a sheet nor a partial sheet.

3.3 Dimensions. Paper in sheet form shall be furnished in the dimensions specified with a tolerance of ± 1/8 inch. Paper in roll form shall be furnished in roll widths of 12, 18, 24, 30, 36, 40, or 48 inches, as specified (see 6.2) with a tolerance of ± 1/8 inch. The diameter of the rolls shall be as specified ± 1/4 inch (see 6.2).

3.4 Physical properties. The kraft wrapping paper shall conform to the applicable requirements of table I, when tested as specified in 4.3.2.

TABLE I. Physical properties

Basis weight (± 5%) 24 X 36 - 500	Average tearing resistance each direction, minimum	Average bursting strength, minimum
Pounds	Grams	Points
	TYPE 1	
Grade A		
35	85	32
60	150	54
80	210	72
Grade B		
30	50	21
40	75	28
50	100	35
60	130	42
70	160	49
80	190	56
	TYPE II	
Grade C		
60	150	54
Grade D		
50	100	35
60	130	42

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3.5 Flame resistance (type II) The paper shall be manufactured or treated to meet the requirements for flaming time, glowing time, and char length specified in table II, when tested as specified in 4.3.2.

TABLE II. Burning characteristics

Maximum flaming time 1/		Maximum glowing time 2/		Maximum char length 3/	
Average of specimens	Individual specimen	Average of specimens	Individual specimen	Average of specimens	Individual specimen
seconds	seconds	seconds	seconds	inches	inches
2.0	2.5	2.0	5.0	4.5	5.5

1/ Flaming time - The time the specimen continues to flame after the burner flame is removed.
2/ Glowing time - The time the specimen continues to glow after it has ceased to flame.
3/ Char length - The height of the void area of the specimen resulting from the breaking out of the material charred during testing measured from the bottom edge of the specimen.

3.5.1 Identification (type II). The words "FIRE RESISTANT TREATED (NOT FOR FOOD SERVICE USE)" in caps, not less than 1 inch high shall be printed in a bright contrasting color, along the entire length of the paper on the roll or individual sheets, as applicable, not closer than 6 inches from the edge on one or both edges of the sheet; the fire resistant treated statement shall be repeated with not more than 12 inches of space between the end of one phrase and the beginning of the next.

3.6 Workmanship. The paper shall be clean and uniform without dirt, dust, spots, shives, lumps, slime spots, foreign or undigested matter, holes, tears, cuts, punctures, splits, wrinkles, or corrugations.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, and standards, unless otherwise excluded, amended, or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.7. Random samples shall be drawn from each lot of the end items for examination of visual, dimensional, roll construction, count, and preparation for delivery defects. The lot shall be expressed in units of rolls or packages of sheets, as applicable, of the same type, grade, basis weight, form and dimension for examination in 4.3.1.1 thru 4.3.1.4, in units of shipping containers for examination in 4.3.1.5, and in units of palletized unit loads for examination in 4.3.1.6.

4.3.1.1 Examination for visual defects. The sample unit for this examination shall be one yard of material the full width of the roll or one sheet, as applicable. No sample unit shall be taken from the first or last convolution of a roll.

Examine	Defect
Form	Not in rolls or sheets, as specified.
Appearance	Not clean; presence of foreign or undigested matter, dirt, dust. Color, finish and formation not uniform throughout.
Workmanship	Shives or slime spots. Any hole, tear, cut, puncture, chafed spot, or scuff mark. Any wrinkle or corrugation. Any splice within a sheet or a partial sheet. Sheets not trimmed square on four sides; ragged, crushed, or uneven edges.

4.3.1.2 Examination for defects in dimensions and identification marking. The sample unit for this examination shall be one roll or sheet, as applicable.

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Examine	Defect
Rolls	
Width	Varies from width specified by more than ± 1/8 inch.
Diameter	Varies from diameter specified by more than ± 1/4 inch.
Core or roll inside diameter (as applicable)	Inside diameter less than 2-3/8 inches.
Identification marking (typeII)	Not marked as specified.
Sheets	
Length	Varies from dimension specified by more than ± 1/8 inch.
width	
Identification marking (type II)	Not marked as specified.

4.3.1.3 Examination for defects in roll construction. The sample unit for this examination shall be one roll.

Examine	Defect
Assembly of roll	End of roll not secured to prevent unwinding. Material not evenly and compactly wound causing soft or uneven edges, or telescoping of the roll. Edges not clean cut; ragged, crushed, nicked, or uneven. Material not wound on a removable shaft or paper core, as specified. Core broken, crushed, collapsed, or mutilated. Length of core varies from width of roll by more than ± 1/4 inch. Either end of core more than 1/4 inch beyond or inside the face of roll.
Plugs	Plugs omitted from one or both ends; do not fit snugly. Hole missing or not centered. Diameter of center hole less than 1 or more than 1-1/8 inch.
Unwinding of roll	Paper slips over core when unwinding. Roll wound unevenly causing wrinkles, sharp creases, or folds within the roll.
Splices	Not evenly and neatly made the entire width of the roll; comes apart during unwinding. More than two splices per roll, more than one splice in 20 consecutive yards of material.

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4.3.1.4 Examination for average count or weight (as applicable). The sample unit for this examination shall be one roll or one package of sheets, as applicable. The lot shall be unacceptable if the average number of sheets per package or the average weight or diameter per roll for all packages or rolls examined, as applicable, is less than the specified or indicated quantity.

4.3.1.5 Examination of preparation for delivery requirements. An examination shall be made to determine that the packaging, packing, and marking complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot shall be the number of shipping containers in the end item inspection lot.

Examine	Defects
Marking (exterior and interior)	Incorrect; incomplete; illegible; omitted, or improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective. Any component not as specified.
Workmanship	Inadequate application of components such as: Incomplete closure of container flaps, loose strapping; improper taping, or inadequate stapling. Bulged or distorted container.
Content	Fiberboard container, or wrap material not as specified. Number of flat cut sheets of paper per interior package is more or less than required.
Weight	Weight of contents per shipping container exceeds requirement.

4.3.1.6 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot shall be the number of palletized unit loads in the end item inspection lot.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.

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Examine	Defect
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified (when applicable). Load not bonded with required straps as specified. Wood cap and double tray cap not used (when applicable).
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.3.1.7 Inspection levels and AQL for examination. The inspection levels for the purpose of determining the sample size and the AQLs expressed as defects per one hundred units shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.1.1	I	2.5
4.3.1.2	S-3	2.5
4.3.1.3	S-3	2.5
4.3.1.4	S-2	N.A.
4.3.1.5	S-1	2.5
4.3.1.6	S-1	6.5

4.3.2 Testing of the end item. The end item shall be tested for the characteristics in table III in accordance with the applicable TAPPI test method. The sample unit for rolls shall be one piece of material 48 inches long by the full width of the roll, and for sheets shall be two full size sheets. No more than one sample unit shall be drawn from any roll or package, as applicable. The lot shall be expressed in units of pounds of paper of one type, grade, form, and basis weight. The sample size shall be drawn in accordance with level S-1 of MIL-STD-105. There shall be no failure to meet the sample unit, lot average or composite requirements. A composite sample shall consist of small, approximately equal, amounts taken from all sample units and combined in a single sample.

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TABLE III. Instructions for testing of the end item

Characteristic	Specification reference		Requirements applicable to		Number of determinations per sample unit	Results reported as	
	Requirement	Test method	Indiv. unit	Lot aver.		Pass or fail	Numerically to nearest
Stock	3.1.1	T 401		X	Average of 2 (on composite)		5 percent
Sizing	3.1.2	T 408	X		1	X	
Basis weight	3.4	T 410		X	1		0.1 pound
Tearing resistance							
Machine direction	3.4	T 414		X	Average of 5		Gram
Cross direction	3.4	T 414		X	Average of 5		Gram
Bursting strength	3.4	T 403		X	Average of 10 (5 on each side)		Pound
Flame resistance							
Flame time (each direction)	3.5	T 461 1/	X		Average of 4 (each direction) 3/		0.2 second
Glowing time (each direction)	3.5	T 461 1/	X		Average of 4 (each direction) 3/		0.2 second
Char length (each (direction)	3.5	T 461 2/	X		Average of 4 (each direction) 3/		0.1 inch

1/ Gas used shall be the synthetic gas mixture with the specified composition.
2/ Use clip weights 12.5 ± 0.5 gram.
3/ No individual specimen shall exceed the specified time or distance, as applicable.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Paper sheets. Five hundred sheets of paper of one description only shall be completely wrapped in the same basis weight kraft paper as that being packaged. The wrap shall be secured with an adhesive, commercially used for this purpose, or 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45. The adhesive shall be applied in such a manner that it does not contact the paper being packaged.

5.1.1.2 Paper rolls. Paper in roll form does not require packaging.

5.1.2 Level B (civil agencies). Paper in sheet and roll form shall be packaged as specified in 5.1.

5.1.3 Level C. Paper shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Paper sheets. Paper sheets of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC or FTC, grade V2s of PPP-B-636. Each fiberboard shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636. The weight of contents of each shipping container shall not exceed 65-pounds.

5.2.1.2 Paper rolls. Paper in roll form and secured as specified in 3.2.1, shall be wrapped overall with waterproofed barrier material conforming to PPP-B-1055, except that when class C-1 is used, two thicknesses shall be required. The wrapper shall be closed at the ends by means of inside and outside headers of the same material as the wrapper. When machine wrapped, inside headers are not required. Seams and outside headers shall be sealed with a water-resistant adhesive conforming to type I or II, grade B, class optional of MMM-A-260, using sufficient adhesive to effect a watertight seal. The width of the continuous seam adhesive strip shall be not less than 3/4-inch.

5.2.2 Level B.

5.2.2.1 Paper sheets. Paper sheets of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC or FTC, type CF (variety SW) or SF, class domestic, grade

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275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. The weight of contents of each shipping container shall not exceed 65-pounds.

5.2.2.1.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.2.2 Paper rolls. Paper in roll form and secured as specified in 3.2.1 shall be wrapped overall with 60-pound minimum basis weight (24 by 36-500) kraft wrapping paper. The wrapper shall be closed at the ends by means of inside and outside headers of the same material as the wrapper. When machine wrapped, inside headers are not required. Seams and outside headers shall be securely sealed with an adhesive commercially used for this purpose or with 2-inch minimum width gummed paper tape conforming to type III, grade C of PPP-T-45.

5.2.3 Level C. Paper, packaged as specified in 5.1, (when applicable), shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Palletization (military requirement).

5.3.1 Paper sheets. When specified (see 6.2), paper sheets of one description only, packed as specified in 5.2.1.1 and 5.2.2.1, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.3.2 Paper rolls. When specified (see 6.2), paper rolls of one description only, packed as specified in 5.2.1.2 and 5.2.2.2, shall be palletized in accordance with load type XIII of MIL-STD-147. Rolls of paper shall be stacked vertically in single or multiple courses (as applicable) with a wood cap over and under the load and a double tray cap (as applicable) between courses. A strap shall be positioned around the load, outside of and at the vertical center of the rolls of paper in each course, and shall be the first strap applied to the load. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L.

5.4 Marking.

5.4.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages, shipping containers, and wrapped rolls shall be marked in accordance with FED-STD-123.

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5.4.2 Military requirements. In addition to any special marking required by the contract or order, interior packages, shipping containers, wrapped rolls and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The kraft wrapping paper covered by this specification is intended for use in general wrapping where chemically neutral or vaporproof barrier properties are not required. Types I and II are not intended for use as a wrapping material for food items. In the event that food wrapping is contemplated (type I only), the paper should be free of reclaimed fiber. Type II paper contains fire resistant properties.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, grade, form, dimensions, roll diameter and width, and basis weight required (see 1.2.1, 3.2, 3.3, and 3.4).
(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather resistant grade fiberboard shipping containers are required for level B shipments (see 5.2.2.1.1).

6.3 Supersession data. This specification includes Military Specification Paper, Kraft, Fire Resistant, MIL-P-43773, dated 19 April 1972, as type II.

6.4 Reclaim fiber sources.

PART I - Post-consumer wastes.

(i) Paper, paperboard and fibrous wastes from factories, retail stores, office buildings, homes, etc., after they have passed through their end-usage as a consumer item, including: used corrugated boxes; old newspapers; old magazines; mixed waste paper; tabulating cards; and used cordage.

(ii) All paper, paperboard, and fiberous wastes that enter and are collected from municipal solid waste.

PART II - Manufacturing, forest residue, and other wastes.

(i) Dry paper and paperboard waste generated after completion of the papermaking process (i.e., those manufacturing operations up to and including the cutting and trimming of the paper machine reel into smaller rolls or rough sheets) including: envelope cuttings, bindery trimmings, and other paper and paperboard waste, resulting from printing, cutting, forming, and other converting operations; bag, box, and carton manufacturing wastes; and butt rolls, mill wrappers, and rejected unused stock.

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(ii) Finished paper and paperboard from obsolete inventories of paper and paperboard manufacturers, merchants, wholesalers, dealers, printers, converters, or others.

(iii) Fibrous by-products of harvesting, manufacturing, extractive, or woodcutting processes, flax straw, linters, bagasse, slash, and other forest residues.

(iv) Wastes generated by the conversion of goods made from fibrous material, i.e., waste rope from cordage manufacture, textile mill waste, and cuttings.

(v) Fibers recovered from waste water which otherwise would enter the waste stream.

Custodians:

Army - GL
Navy - SH
Air Force - 69

Review activities:

Army - MD, ME, SM, SL
Navy - AS, SA
Air Force - 70, 80, 84

User activities:

Army - AT, EL, MI, MU
Navy - CG, MC

Preparing activity:

Army - GL

Civil Agency Interest:

COM - NBS
GSA - FSS
GPO

Project No. 8135-0410

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